

Module 2: The homogeneous solution

2.1 Reading and where this fits

Reading: Chapter 2. Section R.1 of Chapter R: Mechanical Components.

Builds on: Module 1. The split $x(t) = x_h(t) + x_p(t)$, and the general form of an LTIODE.

By the end of this module you should be able to

- write $\Delta(s)$ for an LTIODE of any order and find its roots;
- write $x_h(t)$ for a first- or second-order system in all three root cases, and fix the constants from the initial conditions;
- convert $C_1 \cos \omega t + C_2 \sin \omega t$ into $A \sin(\omega t + \phi)$;
- **sketch** $Ae^{\sigma t} \sin(\omega t + \phi)$, which appears on quizzes and exams;
- decide whether a system is stable, marginally stable or unstable from its roots, and give its time constant and settling time;
- for order three and above, *predict* rather than solve: count and classify the roots, name the dominant pair, and estimate settling and oscillation;
- use Routh–Hurwitz to decide stability, and to find the range of a design parameter that achieves it.

2.2 The homogeneous problem

From Module 1, the solution splits as $x(t) = x_h(t) + x_p(t)$. The homogeneous part is what is left when the input is switched off, so it comes from

$$a_n x^{(n)} + a_{n-1} x^{(n-1)} + \cdots + a_1 \dot{x} + a_0 x = 0, \quad (1)$$

driven only by the initial conditions. First order, then second order, then the general case.

2.3 First-order systems

The system

$$a_1 \dot{x} + a_0 x = f(t), \quad x(0) = x_0.$$

Solve the homogeneous equation

Now do it in a way that will keep working

What does the answer look like?

The sign of the root decides everything, and the three cases are worth seeing before writing them down. **Let's draw them.**

Solution

Time constant and settling time (stable case only)

Solution

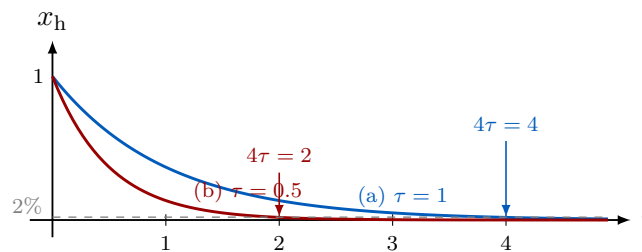
Example: what the time constant tells us

Sketch the homogeneous response of each, with $x(0) = 1$:

$$(a) \quad \dot{x} + x = f(t), \qquad (b) \quad \dot{x} + 2x = f(t).$$

Solution

Both start from $x(0) = 1$, and the difference is easier to see than to state.

**Summary, first order**

2.4 Second-order systems, and $\Delta(s)$ named

$$a_2\ddot{x} + a_1\dot{x} + a_0x = f(t), \quad x(0), \dot{x}(0) \text{ given.}$$

The characteristic equation

The notation, once, for the whole course

s is a *variable*. It is the variable of the characteristic equation $\Delta(s) = 0$, and later of transforms and transfer functions. λ *names values*: the roots of $\Delta(s)$ are written $s = \lambda_1, \dots, \lambda_n$. λ is never a variable.

What kinds of roots are possible?

Case I, two distinct real roots

$$x_h(t) = A_1 e^{\lambda_1 t} + A_2 e^{\lambda_2 t}.$$

Example $\ddot{x} + 3\dot{x} + 2x = f(t), \quad x(0) = 1, \quad \dot{x}(0) = 1.$

Stability and settling time, Case I

Case II, a complex conjugate pair

$\lambda_{1,2} = \sigma \pm j\omega$. Euler's identity $e^{j\theta} = \cos \theta + j \sin \theta$ turns the exponentials into oscillations:

$$e^{(\sigma+j\omega)t} = e^{\sigma t} (\cos \omega t + j \sin \omega t).$$

Example $\ddot{x} + 2\dot{x} + 2x = f(t), \quad x(0) = 1, \quad \dot{x}(0) = 0.$

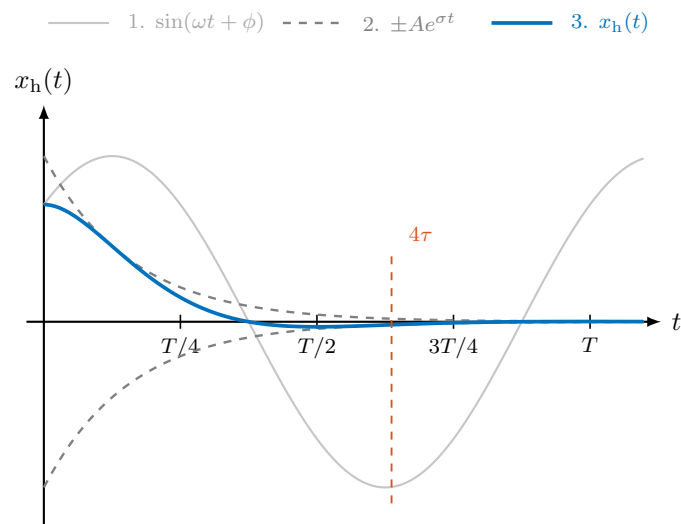
Convention: convert $\cos + \sin$ to a single sine with a phase

Two unknowns either way: C_1, C_2 , or amplitude and phase A, ϕ . The second pair is far easier to sketch.

Solution**Stability and settling time, Case II**

Sketching $Ae^{\sigma t} \sin(\omega t + \phi)$

Steps



The general underdamped shape

Case III, repeated roots

$\lambda_1 = \lambda_2$. The two-exponential form collapses, so the second solution picks up a factor of t :

$$x_h(t) = (A_1 + A_2 t) e^{\lambda_1 t}.$$

Example $\ddot{x} + 2\dot{x} + x = f(t), \quad x(0) = 1, \quad \dot{x}(0) = 0.$

Summary, the three second-order cases

All of it on one chart The whole second-order picture fits on one diagram. With $a_2 = 1$, the pair (a_1, a_0) fixes the roots, so the plane of those two coefficients divides into regions, each with its own kind of response.

2.5 Second-order terminology: ω_n and ζ

Since the dominant pair governs the response of any system, it pays to have precise language for the second-order case. Take a generic second-order LTIOE and divide through so that $a_2 = 1$:

$$\ddot{x} + a_1\dot{x} + a_0x = f(t) \quad \implies \quad \Delta(s) = s^2 + a_1s + a_0.$$

Rewriting the characteristic equation

Example $\ddot{x} + 4\dot{x} + 3x = f(t)$ and $\ddot{x} + \dot{x} + 9x = f(t)$.

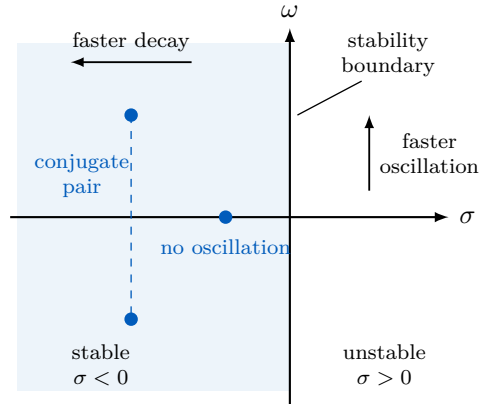
All possible cases:

The same cases seen in the s-plane

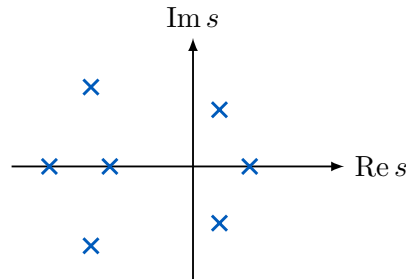
Description

2.6 Roots in the s-plane: the dictionary

Every root is real or one of a complex conjugate pair, so every root is a point $s = \sigma + j\omega$ in the plane. Reading the response off that picture is a skill worth building now. Root locus, in Module 5, is entirely this picture.



The same plane also shows what becomes possible once there are more than two roots.



2.7 The general n th-order system

An n th-order LTIODE has an n th-order characteristic equation,

$$\Delta(s) = a_n s^n + a_{n-1} s^{n-1} + \cdots + a_1 s + a_0,$$

and therefore exactly n roots, each real or part of a conjugate pair. Everything above extends directly.

n roots, n modes

Stability

The slowest root governs

What you are asked to do changes at order three

For first and second order you *solve*. From third order up, solving by hand is not the job and mostly is not possible. The job is to *predict*: how many roots and of what kind, stable or not, which pair dominates, roughly how long it takes to settle and whether it oscillates. Then check by simulation. That is what a practicing engineer does with a high-order model, and it is more useful than a page of algebra.

Example: eight roots, no solving

A system is eighth order, and `roots` returns

$$\lambda_1 = -0.5, \quad \lambda_2 = -6, \quad \lambda_{3,4} = -1 \pm 4j, \quad \lambda_{5,6} = -2 \pm 2j, \quad \lambda_{7,8} = -3 \pm j.$$

Count and classify

Each pair, in ω_n and ζ

Which one governs

The prediction

2.8 Routh–Hurwitz: where the roots are, without finding them

Everything so far needed the roots. For a polynomial with numbers in it, a root finder gives them in one line, and Appendix M shows how. The trouble starts when a coefficient is not yet a number.

The problem Suppose a design parameter c is still to be chosen, and

$$\Delta(s) = s^4 + 10s^3 + cs^2 + 50s + 24.$$

We cannot find roots until c is a number. Guessing values one at a time is a poor plan: we would never know whether a workable c exists at all, nor whether we had found the whole range of them.

Routh–Hurwitz answers the stability question *without* finding a single root, and it answers it for a whole range of a parameter at once.

The necessary condition, and it is free

The Routh table

For $\Delta(s) = a_n s^n + a_{n-1} s^{n-1} + \cdots + a_1 s + a_0$, build a table. The first two rows are the coefficients, dealt alternately. Each later row is built from the two above it.

Solution

The criterion

The number of roots with positive real part equals the number of *sign changes in the first column* of the table.

So: all roots have negative real part if and only if every entry in the first column has the same sign.

Example 1, numeric Is $\Delta(s) = s^3 + 6s^2 + 12s + 8$ stable? [Solution](#)

Example 2, numeric Is $\Delta(s) = s^3 + s^2 + 2s + 8$ stable? [Solution](#)

Example 3, numeric Is $\Delta(s) = s^3 + 2s^2 + s + 6$ stable? [Solution](#)

Example 4, the design question

For what values of $c > 0$ is

$$\Delta(s) = s^3 + 2s^2 + cs + 8$$

stable?

Solution

Axis shifting: settling time, not just stability

In MATLAB

Once the coefficients are numbers, two lines do the whole job:

- `roots([1 2 1 6])` returns the roots;
- `max(real(roots(p)))` is the stability test, since the system is stable exactly when this is negative.

For the axis shift, `poly` rebuilds a polynomial from shifted roots. Appendix M has both.

Routh–Hurwitz earns its keep where MATLAB cannot help: when a coefficient is still a symbol.

2.9 The machine that owns them

Section R.1 gave the element laws and the origin of a stiffness. We apply them here to one object, which we carry from its geometry through to its settling time.

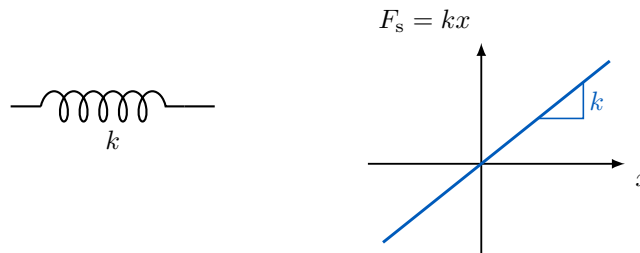
Elastic and dissipative elements

Rigid bodies. The two icons, and the two constants that go with them:



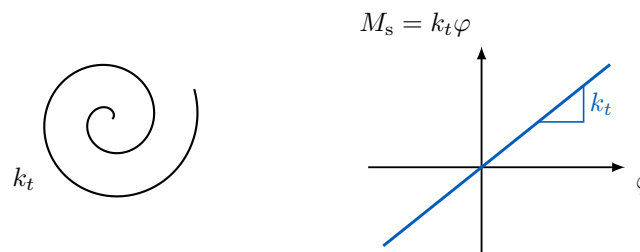
- m [kg], the mass.
- J_G [kg m²], the mass moment of inertia about the center of mass.

Springs. A spring stores energy and gives it back. Its icon is a coil, and its constitutive law says the force is proportional to the deflection:



$$F_s = kx, \quad k \left[\frac{\text{N}}{\text{m}} \right] \text{ the stiffness,} \quad \frac{1}{k} \text{ the compliance.}$$

The torsional spring is the same statement with a torque and an angle in place of a force and a displacement:



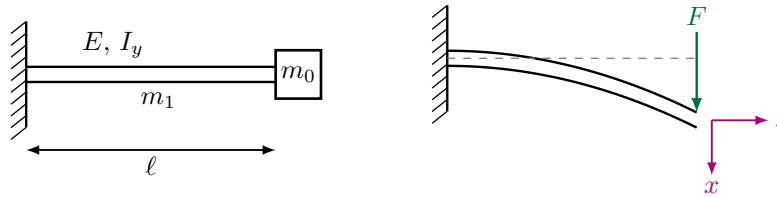
$$M_s = k_t \varphi, \quad k_t [\text{N m}] \text{ the torsional stiffness.}$$

Where a stiffness comes from

Bending: a cantilever with a mass on the end.

Two material and geometric properties enter:

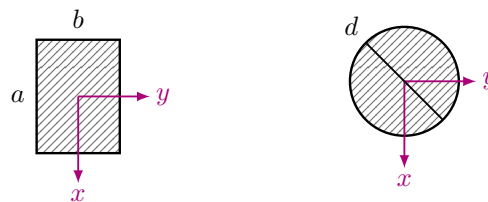
- E , Young's modulus.
- I_y , the area moment of inertia of the cross section, $I_y = \int_{(A)} x^2 dA$.



The tip deflection under a tip load is

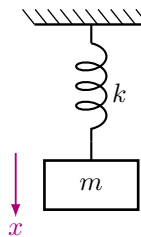
$$x = \frac{\ell^3}{3I_y E} F \quad \Rightarrow \quad \frac{F}{x} = \frac{3I_y E}{\ell^3} =: k.$$

The two cross sections that the note uses:

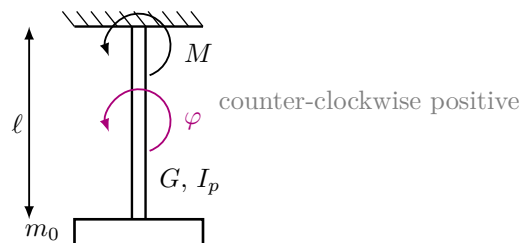


$$I_y = \frac{a^3 b}{12} \quad (\text{rectangle}), \quad I_y = \frac{d^4 \pi}{64} \quad (\text{circle}).$$

The 1 DOF model. The beam becomes a spring of stiffness k and the tip mass becomes the whole mass, $m \approx m_0$:



Torsion: a shaft with a disk on the end.



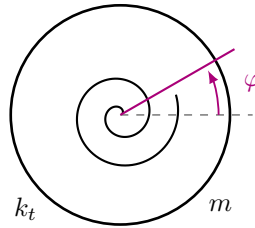
Here the material property is the shear modulus G , which for an isotropic material follows from E and Poisson's ratio ν :

$$G = \frac{E}{2(1 + \nu)},$$

and the geometric one is the polar moment of inertia, $I_p = \int_{(A)} r^2 dA$, which for a circular section is $I_p = \frac{d^4 \pi}{32}$. The twist under an applied torque is

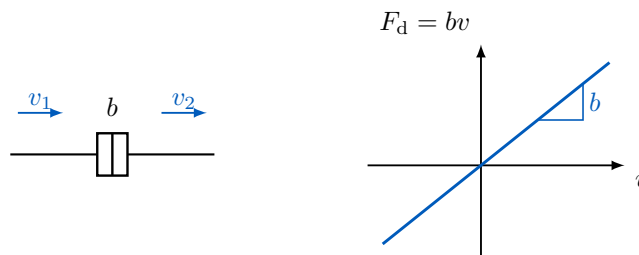
$$\varphi = \frac{\ell}{I_p G} M \quad \Rightarrow \quad \frac{M}{\varphi} = \frac{I_p G}{\ell} =: k_t,$$

and the 1 DOF model is a torsional spring carrying a disk, $m = m_0$:



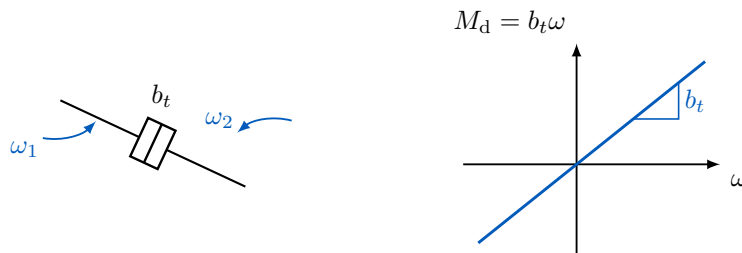
Dampers

A damper dissipates energy. Its force depends not on how far its two ends have moved but on how fast they are moving apart:



$$\underline{v} = \underline{v}_2 - \underline{v}_1, \quad F_d = bv, \quad b \left[\frac{\text{N s}}{\text{m}} \right] \text{ the damping coefficient.}$$

The torsional damper is the same statement with angular velocities:



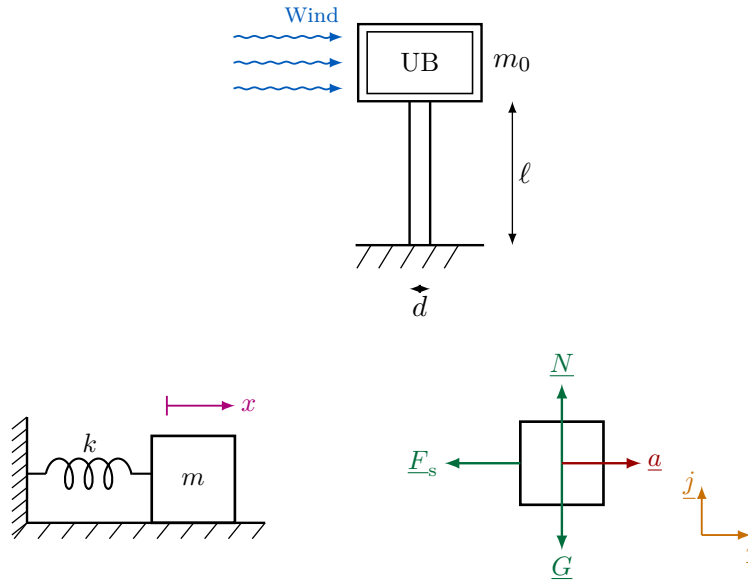
$$\omega = \omega_2 - \omega_1, \quad M_d = b_t \omega, \quad b_t [\text{N m s}] \text{ the torsional damping coefficient.}$$

Example: a slender upright

A thin rod of diameter d and length ℓ , built into the ground at its base and carrying a lumped mass m_0 at the top.

Reduced to a 1 DOF mechanical model, and then to a free body diagram:

Reduce it, then apply Newton's law



Equation of motion

Dividing the $\underline{i}$ equation of the previous page by m :

$$\ddot{x} + \frac{k}{m} x = 0$$

The coefficient carries the units of a frequency squared:

$$\frac{k}{m} \left[\frac{\text{N/m}}{\text{kg}} \right] = \left[\frac{\text{kg} \frac{\text{m}}{\text{s}^2}}{\text{m kg}} \right] = \left[\frac{1}{\text{s}^2} \right],$$

so it is written

$$\omega_n^2 = \frac{k}{m},$$

and ω_n is called the *undamped natural frequency*.

Solution.

Three things are added to the equation of motion of the previous page.

First, the units of ω_n . It is measured in [rad/s], so it is an *angular* natural frequency. Turned into a time, that gives the period:

$$T = \frac{2\pi}{\omega_n} \quad [\text{s}]$$

Second, the velocity. Differentiating the solution,

$$\dot{x}(t) = -\omega_n C_1 \sin(\omega_n t) + \omega_n C_2 \cos(\omega_n t).$$

Third, the initial conditions, which is what turns the two free constants into the two numbers the motion actually starts from:

$$x(0) = C_1 \cdot 1 + C_2 \cdot 0 = x_0 \quad \implies \quad C_1 = x_0,$$

$$\dot{x}(0) = -\omega_n C_1 \cdot 0 + \omega_n C_2 \cdot 1 = v_0 \quad \implies \quad C_2 = \frac{v_0}{\omega_n}.$$

Amplitude and phase

With the constants of the previous page put back,

$$x(t) = x_0 \cos(\omega_n t) + \frac{v_0}{\omega_n} \sin(\omega_n t)$$

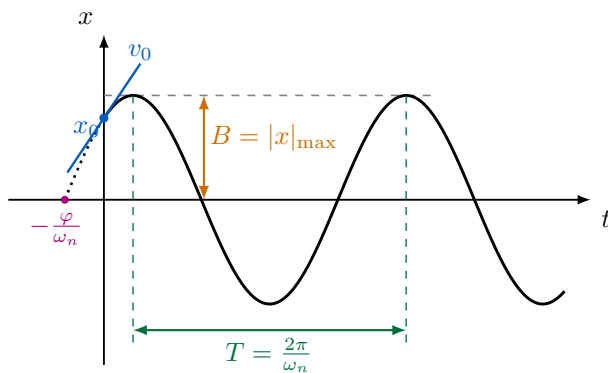
Two sinusoids of the same frequency add up to one, so the same motion can be written

$$x(t) = B \sin(\omega_n t + \varphi) = B \sin \varphi \cos(\omega_n t) + B \cos \varphi \sin(\omega_n t).$$

Matching the two forms term by term,

$$\left. \begin{aligned} B \sin \varphi &= x_0 \\ B \cos \varphi &= \frac{v_0}{\omega_n} \end{aligned} \right\} \implies \quad B = \sqrt{x_0^2 + \frac{v_0^2}{\omega_n^2}} \quad \text{the amplitude,}$$

$$\tan \varphi = \frac{x_0 \omega_n}{v_0} \implies \quad \varphi = \arctan \frac{x_0 \omega_n}{v_0} \quad \text{the phase lag.}$$



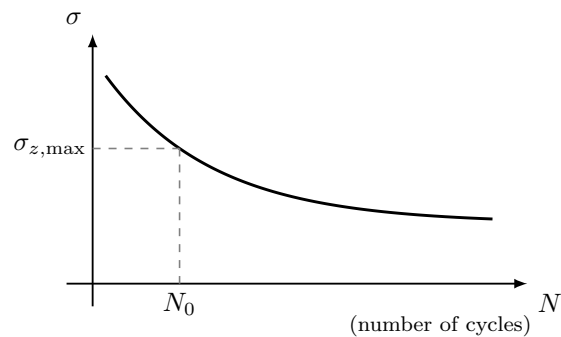
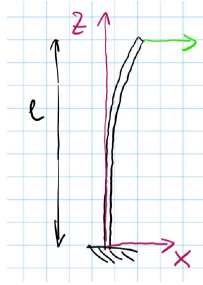
Implementing the results

The largest force the spring ever carries is set by the amplitude:

$$F_{s,\max} = k|x|_{\max} = kB.$$

The largest bending moment is that force acting at the length of the rod:

$$M_{b,\max} \approx F_{s,\max} \ell = kB\ell.$$



The largest stress follows from the bending moment and the section:

$$\sigma_{z,\max} = \frac{M_{b,\max}}{I_y} \cdot \frac{d}{2} = \frac{kB\ell}{d^3\pi} \cdot 32.$$

High cycle fatigue. A stress that is applied once and a stress that is applied a million times are not the same loading. The S–N curve gives the number of cycles the material survives at a given stress:

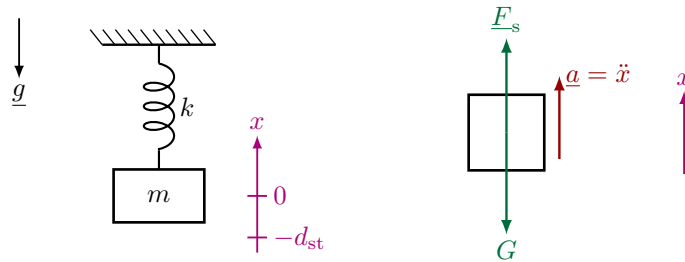
and the period turns that number of cycles into a time:

$$\text{lifetime} = N_0 T = N_0 \frac{2\pi}{\omega_n} \quad [\text{s}] = \cdots \quad [\text{days}].$$

This is the whole argument of the note, end to end. Geometry and material give k ; k and m give ω_n ; ω_n and the initial conditions give B ; B gives the stress; and the stress with the period gives a lifetime in days. Every step is one line, and the dynamics is only the middle of it.

The same system with gravity

A mass hanging on a spring, with x measured upward from the position at which the spring is *unstretched*.



From the free body diagram, $m\ddot{a} = -\underline{F}_s - \underline{G}$, that is

$$m\ddot{x} = -kx - mg \quad / \quad \frac{1}{m}$$

$$\boxed{\ddot{x} + \frac{k}{m}x = -g} \quad \boxed{\omega_n = \sqrt{\frac{k}{m}}}$$

The equation is no longer homogeneous.

Equilibrium. A constant solution $x(t) \equiv -d_{\text{st}}$ has $\dot{x} \equiv 0$ and $\ddot{x} \equiv 0$, so

$$0 - \frac{k}{m}d_{\text{st}} = -g \quad \implies \quad \boxed{d_{\text{st}} = \frac{mg}{k} = \frac{g}{\omega_n^2}}$$

which is the *static deformation*: how far the spring sags under the weight alone.

Solution. The constant $-d_{\text{st}}$ is also the particular solution, so the complete solution is the homogeneous one shifted by it:

$$x(t) = x_{\text{H}}(t) + x_{\text{p}}(t) = C_1 \cos(\omega_n t) + C_2 \sin(\omega_n t) - d_{\text{st}}.$$

Apply the initial conditions.

$$x(0) = C_1 - d_{\text{st}} = x_0 \quad \implies \quad C_1 = x_0 + d_{\text{st}},$$

$$\dot{x}(0) = C_2 \omega_n = v_0 \quad \implies \quad C_2 = \frac{v_0}{\omega_n},$$

$$\boxed{x(t) = (x_0 + d_{\text{st}}) \cos(\omega_n t) + \frac{v_0}{\omega_n} \sin(\omega_n t) - d_{\text{st}}}$$

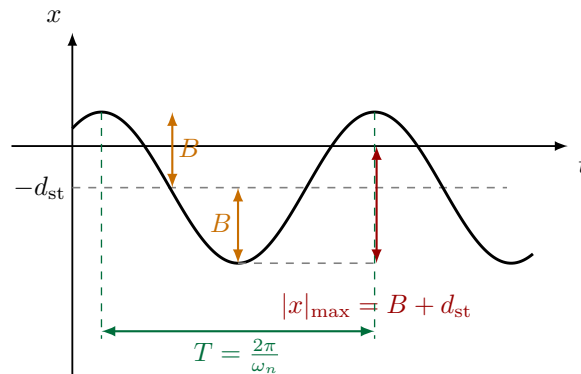
or, in the amplitude and phase form of page 6,

$$\boxed{x(t) = B \sin(\omega_n t + \varphi) - d_{\text{st}}}$$

$$\boxed{B = \sqrt{(x_0 + d_{\text{st}})^2 + \frac{v_0^2}{\omega_n^2}}}$$

$$\boxed{\varphi = \arctan \frac{(x_0 + d_{\text{st}})\omega_n}{v_0}}$$

Both are the page 6 results with x_0 replaced by $x_0 + d_{\text{st}}$, which is the initial displacement measured from the equilibrium rather than from the unstretched position.



The oscillation has the same amplitude B and the same period T as before. Gravity has moved its center line down to $-d_{\text{st}}$ and has changed nothing else about it. What has changed is the largest displacement from the unstretched position:

$$|x|_{\text{max}} = B + d_{\text{st}},$$

and that is the number the force, moment and stress calculation of the implementation page has to be given. Sizing the rod on B alone would leave out the static deflection entirely.